

Carboprost tromethamine: Drug information - UpToDate

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For additional information see "Carboprost tromethamine: Patient drug information"

For abbreviations, symbols, and age group definitions show table

ALERT: US Boxed Warning

Appropriate use:

Carboprost should be used only with strict adherence to recommended dosages.

Experienced physician:

Carboprost should be used by medically trained personnel in a hospital that can provide immediate intensive care and acute surgical facilities.

Brand Names: US

- Hemabate

Brand Names: Canada

- Hemabate

Pharmacologic Category

- Abortifacient;
- Prostaglandin

Dosing: Adult

Refractory postpartum uterine bleeding

Refractory postpartum uterine bleeding: IM: Initial: 250 mcg; if needed, may repeat at 15- to 90-minute intervals; maximum total dose: 2 mg (8 doses)

Termination of pregnancy

Termination of pregnancy: IM: 250 mcg, then 250 mcg at 1.5- to 3.5-hour intervals, depending on uterine response; a 500 mcg dose may be given if uterine response is not adequate after several 250 mcg doses; do not exceed 12 mg total dose or continuous administration for >2 days. **Note:** A 100 mcg test dose may be considered.

Dosing: Kidney Impairment: Adult

Use with caution in patients with a history of renal disease; use is contraindicated in patients with active renal impairment.

Dosing: Liver Impairment: Adult

Use with caution in patients with a history of hepatic disease; use is contraindicated in patients with active hepatic impairment.

Dosing: Older Adult

Refer to adult dosing.

Adverse Reactions

The following adverse drug reactions and incidences are derived from product labeling unless otherwise specified.

Frequency not always defined. Effects due to increased smooth muscle contractility are most common and are generally transient and reversible upon discontinuation of therapy.

Cardiovascular: Chest pain, chest tightness, flushing, hypertension, palpitations, septic shock syncope, tachycardia, vasodepressor syncope

Central nervous system: Anxiety, chills, choking sensation, disturbed sleep, dizziness, drowsiness, dystonia, headache, increased body temperature (may be drug-induced or due to postabortion endometritis), lethargy, nervousness, paresthesia, shivering, vertigo

Dermatologic: Diaphoresis, skin rash

Endocrine & metabolic: Hot flash, increased thirst, thyroid storm

Gastrointestinal: Diarrhea (approximately 67%), vomiting (approximately 67%), nausea (approximately 33%), dysgeusia, epigastric pain, gag reflex, hematemesis, hiccups, retching, xerostomia

Genitourinary: Breast tenderness, cervical perforation (posterior), endometritis (from intrauterine device [IUD]), gynecological pain (dysmenorrhea-like pain), retained placenta (fragment), urinary tract infection, uterine perforation, uterine rupture, sacculization of uterus, uterine hemorrhage

Local: Pain at injection site

Neuromuscular & skeletal: Back pain, leg cramps, myalgia, torticollis, weakness

Ophthalmic: Blepharospasm, blurred vision, eye pain

Otic: Tinnitus

Respiratory: Asthma, bronchospasm, cough, dry throat, dyspnea, epistaxis, hyperventilation, pulmonary edema, respiratory distress, throat irritation (fullness of throat), upper respiratory tract infection, wheezing

<1%, postmarketing, and/or case reports: Hypersensitivity reaction (includes anaphylactic shock, anaphylactoid reaction, anaphylaxis, angioedema)

Contraindications

Hypersensitivity to carboprost tromethamine or any component of the formulation; acute pelvic inflammatory disease; active cardiac, pulmonary, renal, or hepatic disease.

Warnings/Precautions

Concerns related to adverse effects:

- **Fever:** Transient fever may be observed with treatment and is believed to be due to carboprost's effect on hypothalamic thermoregulation; use caution to distinguish between fever induced by the drug and fever associated with postabortion endometritis.
- **Gastrointestinal effects:** Pretreatment or concomitant use of antiemetic and antidiarrheal agents is recommended to decrease the incidence of gastrointestinal effects.
- **Hypertension:** Increased blood pressure may be observed with treatment; degree of hypertension observed is generally moderate and does not require treatment.

Disease-related concerns:

- Anemia: Use with caution in patients with anemia.
- Asthma: Use with caution in patients with a history of asthma; therapy may cause transient bronchoconstriction.
- Cardiovascular disease: Use with caution in patients with cardiovascular disease, including hypotension or hypertension.
- Compromised uteri: Use oxytocic agents with caution in patients with compromised (scarred) uterus.
- Diabetes: Use with caution in patients with diabetes mellitus.
- Hepatic impairment: Use with caution in patients with hepatic disease, including jaundice.
- Renal impairment: Use with caution in patients with renal disease.
- Seizures: Use with caution in patients with a history of seizure disorder.

Dosage form specific issues:

- Benzyl alcohol and derivatives: Some dosage forms may contain benzyl alcohol; large amounts of benzyl alcohol (≥ 99 mg/kg/day) have been associated with a potentially fatal toxicity ("gasping syndrome") in neonates; the "gasping syndrome" consists of metabolic acidosis, respiratory distress, gasping respirations, CNS dysfunction (including convulsions, intracranial hemorrhage), hypotension and cardiovascular collapse (AAP ["Inactive" 1997]; CDC 1982); some data suggests that benzoate displaces bilirubin from protein binding sites (Ahlfors 2001); avoid or use dosage forms containing benzyl alcohol with caution in neonates. See manufacturer's labeling.

Other warnings/precautions:

- Potent oxytocic agent: **[US Boxed Warning] Potent oxytocic agent; use only with strict adherence to recommended dosing. Carboprost should be used only by medically-trained personnel in a hospital which can provide immediate intensive care and acute surgical facilities.**

Dosage Forms: US

Excipient information presented when available (limited, particularly for generics); consult specific product labeling.

Solution, Intramuscular [strength expressed as base]:

Hemabate: 250 mcg/mL (1 mL) [contains benzyl alcohol]

Generic: 250 mcg/mL (1 mL)

Solution Prefilled Syringe, Intramuscular:

Generic: 250 mcg/mL (1 mL)

Generic Equivalent Available: US

Yes

Pricing: US

Solution (Carboprost Tromethamine Intramuscular)

250 mcg/mL (per mL): \$41.76 - \$382.79

Solution (Hemabate Intramuscular)

250 mcg/mL (per mL): \$168.00

Solution Prefilled Syringe (Carboprost Tromethamine Intramuscular)

250 mcg/mL (per mL): \$150.00

Disclaimer: A representative AWP (Average Wholesale Price) price or price range is provided as reference price only. A range is provided when more than one manufacturer's AWP price is available and uses the low and high price reported by the manufacturers to determine the range. The pricing data should be used for benchmarking purposes only, and as such should not be used alone to set or adjudicate any prices for reimbursement or purchasing functions or considered to be an exact price for a single product and/or manufacturer. Medi-Span expressly disclaims all warranties of any kind or nature, whether express or implied, and assumes no liability with respect to accuracy of price or price range data published in its solutions. In no event shall Medi-Span be liable for special, indirect, incidental, or consequential damages arising from use of price or price range data. Pricing data is updated monthly.

Dosage Forms: Canada

Excipient information presented when available (limited, particularly for generics); consult specific product labeling.

Solution, Intramuscular [strength expressed as base]:

Hemabate: 250 mcg/mL (1 mL) [contains benzyl alcohol]

Generic: 250 mcg/mL (1 mL)

Administration: Adult

IM: Administer deep IM (use a tuberculin syringe for termination of pregnancy); rotate site if repeat injections are required. Do not inject IV

Hazardous Drugs Handling Considerations

This medication is not on the NIOSH (2024) list; however, it may meet the criteria for a hazardous drug. Carboprost may cause reproductive toxicity and has a structural/toxicity profile similar to existing hazardous agents.

Use appropriate precautions for receiving, handling, storage, preparation, dispensing, transporting, administration, and disposal. Follow NIOSH and USP 800 recommendations **and** institution-specific policies/procedures for appropriate containment strategy (NIOSH 2023; NIOSH 2024; USP-NF 2020).

Note: Facilities may perform risk assessment of some hazardous drugs to determine if appropriate for alternative handling and containment strategies (USP-NF 2020). Refer to institution-specific handling policies/procedures.

Storage/Stability

Store under refrigeration at 2°C to 8°C (36°F to 46°F).

Use: Labeled Indications

Termination of pregnancy: For aborting pregnancy between week 13 and 20 of gestation as calculated from the first day of the last normal menstrual period and in the following conditions related to second trimester abortion: Failure of expulsion of the fetus during the course of treatment by another method; premature rupture of membranes in intrauterine methods with loss of drug and insufficient or absent uterine activity; requirement of a repeat intrauterine instillation of drug for expulsion of the fetus; inadvertent or spontaneous rupture of membranes in the presence of a previable fetus and absence of adequate activity for expulsion.

Refractory postpartum uterine hemorrhage: Treatment of postpartum hemorrhage due to uterine atony that has not responded to conventional methods of management. Prior treatment should include the use of intravenously (IV) administered oxytocin, manipulative techniques such as uterine massage and, unless contraindicated, intramuscular ergot preparations.

Metabolism/Transport Effects

None known.

Drug Interactions

(For additional information: Launch drug interactions program)

Note: Interacting drugs may **not be individually listed below** if they are part of a group interaction (eg, individual drugs within “CYP3A4 Inducers [Strong]” are NOT listed). For a complete list of drug interactions by individual drug name and detailed management recommendations, use the drug interactions program by clicking on the “Launch drug interactions program” link above.

Dinoprostone: May increase adverse/toxic effects of Oxytotic Agents. Specifically, the oxytotic effects may be increased. Management: Consider alternatives when possible, as product labeling states that the use of dinoprostone in combination with oxytotic agents is not recommended. If combined, the risk for uterine overstimulation may be increased. *Risk D: Consider Therapy Modification*

Methylergonovine: Adverse/toxic effects of Methylergonovine and Prostaglandins may increase with coadministration. *Risk C: Monitor*

MiSOPROStol: May increase adverse/toxic effects of Oxytotic Agents. Specifically, the oxytotic effects may be increased. Management: The concomitant use of misoprostol with other oxytotic agents is not recommended. If sequential use of oxytocin is necessary, oxytocin should be given at least 4 hours after misoprostol. *Risk D: Consider Therapy Modification*

Oxytotic Agents: Carboprost Tromethamine may increase adverse/toxic effects of Oxytotic Agents. Specifically, oxytotic effects may be enhanced. Management: Consider alternatives when possible, as product labeling states that the use of carboprost tromethamine in combination with oxytotic agents is not recommended. If combined, the risk for uterine overstimulation may be increased. *Risk D: Consider Therapy Modification*

Sulprostone: May increase adverse/toxic effects of Oxytotic Agents. Management: Consider alternatives, as the use of sulprostone in combination with oxytotic agents is not recommended. If combined, the risk for uterine overstimulation may be increased. This does not apply when sulprostone is used for postpartum atonic bleeding *Risk D: Consider Therapy Modification*

Pregnancy Considerations

When used for termination of pregnancy, carboprost is not considered fetocidal, but is used to terminate pregnancy due to its ability to stimulate uterine contractions; use is not indicated if the fetus has reached a stage of viability in utero. Complete termination of pregnancy may not be induced in ~20% of cases and should therefore be completed in another way.

Breastfeeding Considerations

It is not known if carboprost is excreted in breast milk.

Monitoring Parameters

Termination of pregnancy: Confirmation of fetal death; cervical exam after termination of pregnancy

Mechanism of Action

Carboprost is an analog of naturally occurring prostaglandin F₂ alpha (dinoprost); carboprost stimulates uterine contractility which usually results in expulsion of the products of conception and is used to induce abortion between 13-20 weeks of pregnancy. When used postpartum, hemostasis at the placentation site is achieved through the myometrial contractions produced by carboprost.

Pharmacokinetics (Adult Data Unless Noted)

Time to peak, serum: IM: 30 minutes

Patient Counseling Points

(For additional information see “Carboprost tromethamine: Patient drug information”)

What is this drug used for?

- It is used to end a pregnancy.
- It is used to stop or treat bleeding that happens after a birth or an abortion.
- It may be given to you for other reasons. Talk with the doctor.

All drugs may cause side effects. However, many people have no side effects or only have minor side effects. Call your doctor or get medical help if any of these side effects or any other side effects bother you or do not go away:

- Nausea
- Vomiting
- Diarrhea

WARNING/CAUTION: Even though it may be rare, some people may have very bad and sometimes deadly side effects when taking a drug. Tell your doctor or get medical help right away if you have any of the following signs or symptoms that may be related to a very bad side effect:

• Signs of an allergic reaction, like rash; hives; itching; red, swollen, blistered, or peeling skin with or without fever; wheezing; tightness in the chest or throat; trouble breathing, swallowing, or talking; unusual hoarseness; or swelling of the mouth, face, lips, tongue, or throat.

Note: This is not a comprehensive list of all side effects. Talk to your doctor if you have questions.

Consumer Information Use and Disclaimer: This information should not be used to decide whether or not to take this medicine or any other medicine. Only the healthcare provider has the knowledge and training to decide which medicines are right for a specific patient. This information does not endorse any medicine as safe, effective, or approved for treating any patient or health condition. This is only a limited summary of general information about the medicine's uses from the patient education leaflet and is not intended to be comprehensive. This limited summary does NOT include all information available about the possible uses, directions, warnings, precautions, interactions, adverse effects, or risks that may apply to this medicine. This information is not intended to provide medical advice, diagnosis or treatment and does not replace information you receive from the healthcare provider. For a more detailed summary of information about the risks and benefits of using this medicine, please speak with your healthcare provider and review the entire patient education leaflet.

Brand Names: International

International Brand Names by Country

For country code abbreviations (show table)

- (AE) United Arab Emirates: Hemabate;
- (AU) Australia: Carboprost reach;
- (CN) China: An lie ke | Hemabate;
- (CZ) Czech Republic: Prostin 15m;
- (GB) United Kingdom: Hemabate;
- (HK) Hong Kong: Hemabate;
- (IN) India: Caboprost | Deviprost | Evacarb | Femiprost;
- (KE) Kenya: Endoprost;
- (KW) Kuwait: Hemabate;
- (LB) Lebanon: Prostin 15m;
- (NL) Netherlands: Prostin 15m;
- (NO) Norway: Prostin 15M orifarm;
- (NZ) New Zealand: Prostin 15m;
- (PH) Philippines: Endoprost | Evacarb | Evaprost | Prostodin;
- (PL) Poland: Prostinfenem;

- (PR) Puerto Rico: Carboprost tromethamine | Hemabate;
- (QA) Qatar: Hemabate;
- (SA) Saudi Arabia: Hemabate;
- (SG) Singapore: Hemabate;
- (SI) Slovenia: Karboprost | Questtran;
- (TH) Thailand: Hemabate;
- (TW) Taiwan: Hemabate;
- (UG) Uganda: Endoprost

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REFERENCES

1. Ahlfors CE. Benzyl alcohol, kernicterus, and unbound bilirubin. *J Pediatr*. 2001;139(2):317-319. [PubMed 11487763]
2. Centers for Disease Control (CDC). Neonatal deaths associated with use of benzyl alcohol—United States. *MMWR Morb Mortal Wkly Rep*. 1982;31(22):290-291. <http://www.cdc.gov/mmwr/preview/mmwr> [PubMed 6810084]
3. Hemabate (carboprost tromethamine injection) [prescribing information]. New York, NY: Pfizer; March 2025.
4. Hodson L, Ovesen J, Couch J, et al; US Centers for Disease Control and Prevention, National Institute for Occupational Safety and Health. Managing hazardous drug exposures: information for healthcare settings, 2023. <https://doi.org/10.26616/NIOSH PUB2023130>. Updated April 2023. Accessed December 27, 2024.
5. "Inactive" ingredients in pharmaceutical products: update (subject review). American Academy of Pediatrics (AAP) Committee on Drugs. *Pediatrics*. 1997;99(2):268-278. [PubMed 9024461]
6. Ippoliti C, Przepiorka D, Mehra R, et al, "Intravesicular Carboprost for the Treatment of Hemorrhagic Cystitis After Marrow Transplantation," *Urology*, 1995, 46(6):811-5. [PubMed 7502421]
7. Levine LA, Jarrard DF. Treatment of cyclophosphamide-induced hemorrhagic cystitis with intravesical carboprost tromethamine. *J Urol*. 1993;149(4):719-723. doi:10.1016/s0022-5347(17)36192-x [PubMed 8455230]
8. Ovesen JL, Sammons D, Connor TH, et al; US Centers for Disease Control and Prevention, National Institute for Occupational Safety and Health. NIOSH list of hazardous drugs in healthcare settings, 2024. <https://doi.org/10.26616/NIOSH PUB2025103>. Updated December 18, 2024. Accessed December 20, 2024.
9. United States Pharmacopeia. <800> Hazardous Drugs—Handling in Healthcare Settings. In: USP-NF. United States Pharmacopeia; July 1, 2020. Accessed January 16, 2025. doi:10.31003/USPNF_M7808_07_01
10. US Department of Health and Human Services (HHS) Panel on Opportunistic Infections in Adults and Adolescents with HIV. Guidelines for the prevention and treatment of opportunistic infections in adults and adolescents with HIV. <https://clinicalinfo.hiv.gov/en/guidelines/adult-and-adolescent-opportunistic-infection>. Updated May 2, 2024. Accessed June 17, 2024.
11. West NJ, "Prevention and Treatment of Hemorrhagic Cystitis," *Pharmacotherapy*, 1997, 17(4):696-706. [PubMed 9250547]

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